

Introduction Linux presentation

Security is a cornerstone in the management of any IT infrastructure, and Linux systems are no exception. As one of the most widely used operating systems for servers, cloud environments, and even embedded devices, ensuring the integrity and safety of Linux systems is critical. With the increasing complexity of cyber threats, Linux servers are frequently targeted by attackers looking to exploit vulnerabilities.

To defend against these threats, Linux provides a variety of built-in security mechanisms, such as Firewalld, SELinux, and SSH.

In this presentation, we will explore these essential security tools and how they function. We'll discuss their configurations, best practices, and why these tools are fundamental in protecting your Linux environment against evolving security threats.

SELinux

Overview and history:

SELinux (Security-Enhanced Linux) was developed by the National Security Agency (NSA) in the early 2000s as part of a project to improve Linux system security. It was released under an open-source license and integrated into mainstream Linux distributions, starting with Red Hat Enterprise Linux 4. Over time, it has become a critical component in securing systems against exploits, misconfigurations, and privilege escalations, making it essential for enterprise-grade Linux security.

SELinux is a powerful Linux security module that uses Mandatory Access Control (MAC) to enforce strict policies. Unlike traditional controls based on user permissions (DAC), SELinux confines processes and limits their access to files, ports, and resources using security labels and contexts.

Understanding SELinux (Security-Enhanced Linux):

SELinux is a security feature in Linux that controls what actions processes can perform on files and resources. It works by assigning every file, process, and system object a security context that looks like this:

user:role:type:level.

These labels define the permissions for each object, and SELinux enforces these permissions using policies—pre-defined rules that specify what actions are allowed.

How SELinux Works

When a process (like a program) tries to access a file or resource:

SELinux checks its policy: It looks at the process's and the file's security labels.

Access is allowed or denied: If the action matches the policy, it's allowed; otherwise, it's blocked.

For example:

A web server can only read files labeled `httpd_sys_content_t`.

If the web server tries to write to these files, SELinux blocks the action.

Modes of SELinux

SELinux operates in three modes:

Enforcing: Actively blocks any unauthorized actions based on its policies.

This is the most secure mode.

Permissive: Logs unauthorized actions but doesn't block them.

Useful for debugging and testing policies.

Disabled: Turns SELinux off entirely.

No security enforcement happens in this mode.

Why SELinux Is Important

Logs for troubleshooting: Denied access attempts are logged in `/var/log/audit/audit.log`, helping administrators understand and fix issues.

Reduces risks: By strictly enforcing policies, SELinux minimizes unauthorized access, prevents privilege escalation, and limits damage from compromised processes.

To activate the mode : enforcing (où SELinux applique strictement ses politiques) :

```
sudo setenforce 1 // enforcing
```

To activate the mode : permissive (où SELinux permet les actions mais enregistre les violations sans les bloquer) :

```
sudo setenforce 0 // permissive
```

To show the status de SELinux (mode actif, politique appliquée, etc.) :

sudo getenforce

Ou pour obtenir des détails plus complets sur la configuration de SELinux :

sudo sestatus

Part Summary:

SELinux is a security module integrated into the Linux kernel that enforces mandatory access control (MAC) policies. It operates in 3 modes: permissive, enforcing and disabled

SELinux assigns a security context to every file, directory, process, and user, controlling their interactions.